Python Practice for Data Analyst Preparation

30 practical coding questions | 3 difficulty levels

These 30 exercises are based on the ten notebooks reviewed. They build on variables, core data types, strings and indexing, lists, tuples, sets, dictionaries, type conversion, `if`/`elif`/`else`, `for` and `while` loops, nested loops, functions, `*args`/`**kwargs`, and the BMI/file-sorting examples. All datasets below can be entered directly as Python values; no external libraries are required.

Beginner — core skills

1. Sales total — Given prices = [12.50, 8.00, 15.75, 4.25], calculate and print the total sales using a loop and a running total.

2. Type cleanup — Convert quantities = ['3', '10', '2', '5'] to integers, then print the total quantity and the converted list.

3. Revenue per item — Given units = [4, 2, 5] and unit_prices = [3.5, 10.0, 2.25], use a loop to calculate each line's revenue and print the overall revenue.

4. Pass/fail classification — For scores = [88, 42, 73, 59, 96], print each score followed by Pass for 60 or higher, otherwise Needs review.

5. Customer name formatting — Given names = [' maya rai ', 'ARUN SHAH', 'sita thapa '], remove surrounding spaces and print each name with consistent title-style capitalization.

6. Find a product — Given products = ['tea', 'coffee', 'rice', 'tea', 'oil'], count how many times tea appears using a loop and print the result.

7. Unique regions — Given regions = ['East', 'West', 'East', 'North', 'West', 'South'], convert the values to a set and print the unique regions and their count.

8. Simple record — Create a dictionary for one sale with keys product, quantity, and unit_price. Calculate and add a revenue value, then print every key and value.

9. BMI function — Write a function that accepts weight in pounds and height in inches, calculates BMI using the formula in your notebook, and returns the value. Call it with two sample inputs and print the result.

10. Daily totals — Given daily_sales = [120, 95, 150, 80, 110], use a loop to calculate the total, average, highest value, and lowest value. Print all four results.

Intermediate — combine concepts

11. Sales by product — Given sales = [{'product': 'Tea', 'qty': 3, 'price': 2.5}, {'product': 'Coffee', 'qty': 2, 'price': 5}, {'product': 'Tea', 'qty': 4, 'price': 2.5}], calculate total revenue for each product and print a dictionary of product totals.

12. Discount bands — Write a function that takes a purchase amount and returns a discount rate: 0% below 50, 5% from 50 through 99.99, and 10% at 100 or more. Apply it to [35, 75, 120] and print each amount and its discounted price.

13. Data quality audit — Given ages = ['25', '', 'unknown', '31', ' ', '40'], loop through the values, convert valid entries to integers, and count valid, blank, and invalid entries separately. Print the cleaned values and counts.

14. Category counts — For categories = ['Food', 'Travel', 'Food', 'Office', 'Travel', 'Food'], build and print a dictionary containing each category and its frequency.

15. Compare customer lists — Given month_a = {'Asha', 'Ben', 'Carlos', 'Devi'} and month_b = {'Ben', 'Devi', 'Eli'}, print customers retained, newly appearing, and no longer appearing, using set operations.

16. Average by department — Given records = [{'department':'Sales','score':80}, {'department':'Support','score':90}, {'department':'Sales','score':70}, {'department':'Support','score':75}], calculate each department's average score. Use dictionaries and loops; do not hard-code the final averages.

17. Reusable summary function — Write a function that accepts a list of numeric values and returns a dictionary with count, total, average, minimum, and maximum. Use it to summarize two different lists.

18. Keyword-driven report — Write a function accepting kwargs that prints a compact report from fields such as name, region, and revenue. Call it with two different records and handle whichever provided fields exist.

19. Top sales day — Given sales = [120, 180, 95, 180, 140], find the maximum and print every day index (starting at 1) that reached it. Use loops and conditions, without using max() or .index().

20. File sorter plan — Given files = ['jan.xlsx', 'photo.jpg', 'notes.pdf', 'clip.mp4', 'report.docx', 'data.csv', 'logo.jpg'], use conditions and a loop to group filenames into dictionaries/lists for image, video, spreadsheet, document, PDF, and other categories. This exercise models the classification part of your file-sorter notebook; only print the plan, do not move files.

Advanced — analyst-style mini problems

21. Multi-metric sales report — Given orders = [{'region':'East','product':'Tea','qty':'4','price':'2.5'}, {'region':'West','product':'Coffee','qty':'2','price':'6'}, {'region':'East','product':'Coffee','qty':'3','price':'6'}, {'region':'West','product':'Tea','qty':'5','price':'2.5'}], convert numeric strings, calculate row revenue, total revenue by region, total revenue by product, and the overall average order value. Print a clear report.

22. Flag unusual values — Given daily_counts = [18, 20, 19, -2, 21, 250, 20], calculate the average of nonnegative values, then print values more than twice that average or below zero with their positions. Explain in a comment why checking anomalous values before summarizing matters.

23. Survey response cleanup — Given responses = ['East', ' east ', 'WEST', '', 'N/A', 'East', 'west'], normalize valid regions to consistent capitalization, exclude blanks and N/A, and produce a frequency dictionary plus a set of distinct cleaned regions.

24. Customer transaction summary — Given transactions = [{'customer':'Mina','amount':40}, {'customer':'Raj','amount':75}, {'customer':'Mina','amount':25}, {'customer':'Raj','amount':10}, {'customer':'Lee','amount':90}], calculate total spend and transaction count per customer, then print customers ordered by total spend using a simple repeated-selection loop (no sorting libraries required).

25. Inventory exception report — Given inventory = [{'item':'Pen','stock':'12','reorder_at':10}, {'item':'Paper','stock':'4','reorder_at':8}, {'item':'Folder','stock':'unknown','reorder_at':5}, {'item':'Stapler','stock':'0','reorder_at':2}], convert valid stock values, list items at or below reorder level, and separately report invalid stock entries.

26. Compare two periods — Given before = {'Tea': 100, 'Coffee': 80, 'Rice': 50} and after = {'Tea': 120, 'Coffee': 60, 'Oil': 40}, identify added, removed, and shared products. For shared products calculate absolute and percentage change; handle zero baselines safely.

27. Reusable KPI calculator — Write a function that accepts any number of revenue values using *args and a target as a keyword argument. Return total revenue, average revenue, number of values meeting/exceeding target, and whether total revenue meets an overall goal. Demonstrate it with at least two calls.

28. Nested regional report — Given data = [{'region':'East','month':'Jan','sales':100}, {'region':'East','month':'Feb','sales':130}, {'region':'West','month':'Jan','sales':90}, {'region':'West','month':'Feb','sales':110}, {'region':'East','month':'Jan','sales':25}], build a nested dictionary of sales by region and month, correctly accumulating duplicate rows. Print each region's monthly values and total.

29. BMI cohort analysis — Reuse the BMI formula and category thresholds from your notebook. For people = [{'name':'A','weight':150,'height':65}, {'name':'B','weight':210,'height':70}, {'name':'C','weight':120,'height':64}, {'name':'D','weight':180,'height':68}], calculate each BMI, assign a category with a function, reject nonpositive inputs, and report category counts and average BMI.

30. Analyst's end-to-end mini project — Start with raw_orders = [{'region':' east ','item':'Tea','quantity':'3','unit_price':'2.5'}, {'region':'WEST','item':'Coffee','quantity':'2','unit_price':'6'}, {'region':'East','item':'Tea','quantity':'bad','unit_price':'2.5'}, {'region':'West','item':'Rice','quantity':'5','unit_price':'1.8'}, {'region':'','item':'Coffee','quantity':'1','unit_price':'6'}]. Clean region and item text, convert numeric fields, separate valid rows from rejected rows, calculate revenue for valid orders, and report total revenue by region and item plus the number of rejected rows. Organize the solution into small functions and print a readable summary.

Suggested practice routine

For each question, first write down the expected inputs and outputs. Solve it using only the concepts introduced up to that level. Then test normal cases and edge cases such as empty lists, invalid numeric strings, duplicate records, and zero values. The advanced exercises combine familiar Python constructs into analyst workflows; they do not require pandas or other libraries.